Module:5

Design of Sequential Logic Circuits

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Module:5 Design of Sequential Logic Circuits

- Latches, Flip-Flops - SR, D, JK & T,
- Buffer Registers, Shift Registers - SISO, SIPO, PISO, PIPO,
- Design of synchronous sequential circuits: state table and state diagrams,
- Design of counters: Modulo-n, Johnson, Ring, Up/Down, Asynchronous counter.
- Modeling of sequential logic circuits using Verilog HDL.

Introduction

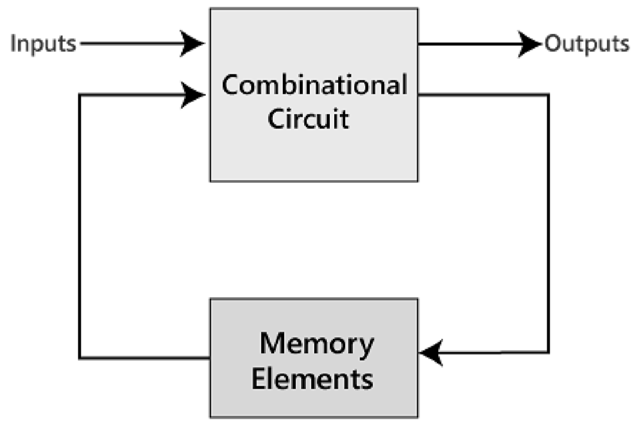
- Sequential logic circuits are those, whose output depends on both present and past values of their inputs.

- Sequential circuits act as storage elements and have memory.

- They can store, retain, and then retrieve information when needed at a later time.

- It can be considered as combinational circuit with feedback.

- It uses a memory element like flip – flops as feedback circuit in order to store past values. The block diagram of a sequential logic is shown below.



Combinational Circuits vs Sequential Circuits

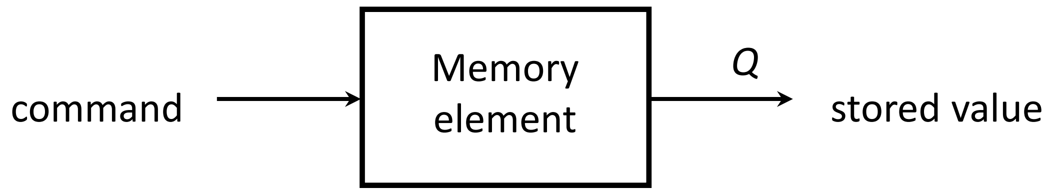
COMBINATIONAL CIRCUITS	SEQUENTIAL CIRCUITS
Output depends only on the present value of the inputs.	Output depends on both the present and previous state values of the inputs.
These circuits will not have any memory as their outputs change with the change in the input value.	Sequential circuits have some sort of memory as their output changes according to the previous and present values.
There are no feedbacks involved.	In a sequential circuit the outputs are connected to it as a feedback path.
Used in basic Boolean operations.	Used in the designing of memory devices.
Implemented in: Half adder circuit, full adder circuit, multiplexers, demultiplexers, decoders and encoders.	Implemented in: RAM, Registers, counters and other state retaining machines.

Introduction

- There are two types of sequential circuits:
  - *synchronous*: outputs change only at specific time
  - *asynchronous*: outputs change at any time
- *Multivibrator*: a class of sequential circuits. They can be:
  - *bistable* (2 stable states)
  - *monostable* (1 stable state)
  - *astable* (no stable state)
- Bistable logic devices: *latches* and *flip-flops*.
- Latches and flip-flops differ in the method used for changing their state.

Memory Elements

- **Memory element**: a device which can remember value indefinitely, or change value on command from its inputs.



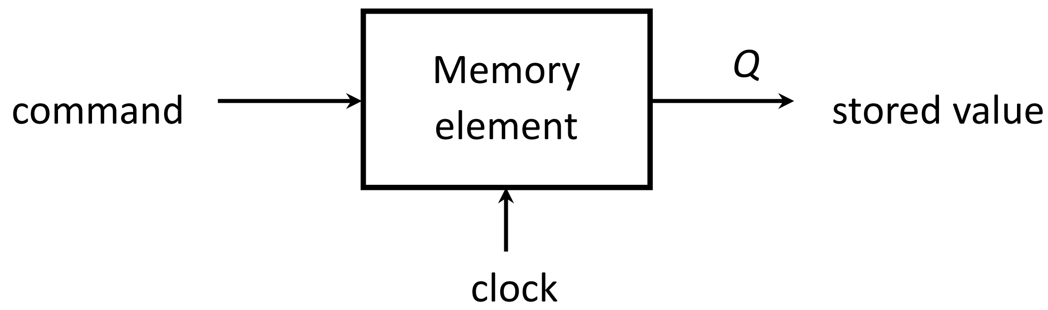
- Characteristic table:

Command (at time *t*)	*Q(t)*	*Q(t+1)*
Set	X	1
Reset	X	0
Memorise / No Change	0	0
	1	1

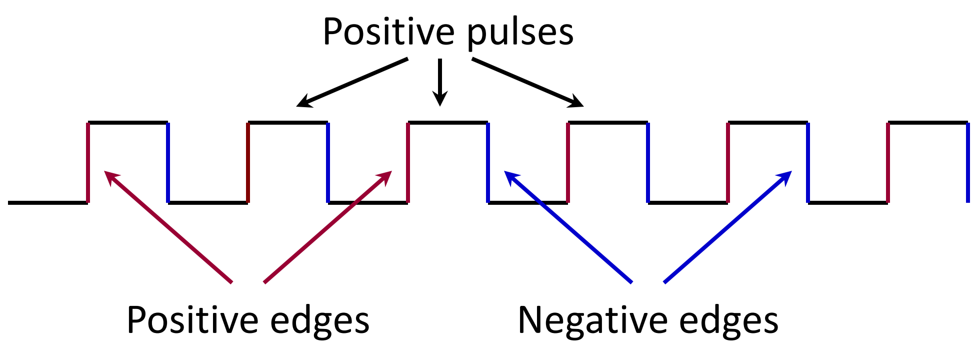
Q(t): current state
Q(t+1) or Q^+: next state

Memory Elements

- Memory element with clock - memory elements that change state on clock signals.



- Clock is usually a square wave.



Memory Elements

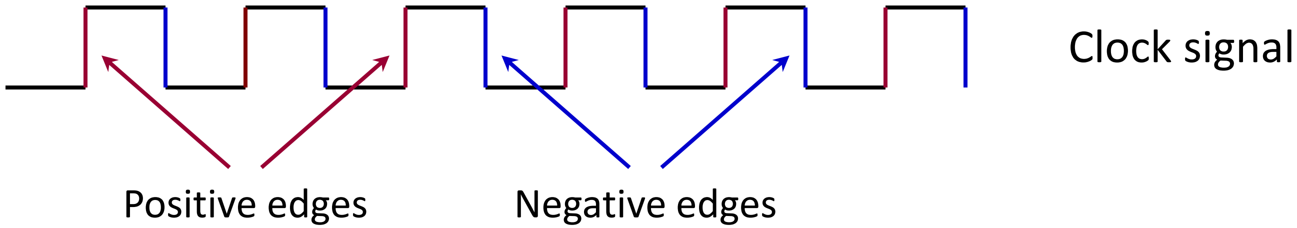
- Two types of triggering/activation:
  - ❖ pulse-triggered
  - ❖ edge-triggered
- Pulse-triggered
  - ❖ **Latches:** Storage elements that operate with signal levels
  - ❖ ON = 1, OFF = 0
- Edge-triggered
  - ❖ **Flip-flops:** Storage elements controlled by a clock transition
  - ❖ positive edge-triggered (ON = from 0 to 1; OFF = other time)
  - ❖ negative edge-triggered (ON = from 1 to 0; OFF = other time)

Edge-Triggered Flip-flops

- *Flip-flops*: synchronous bistable devices

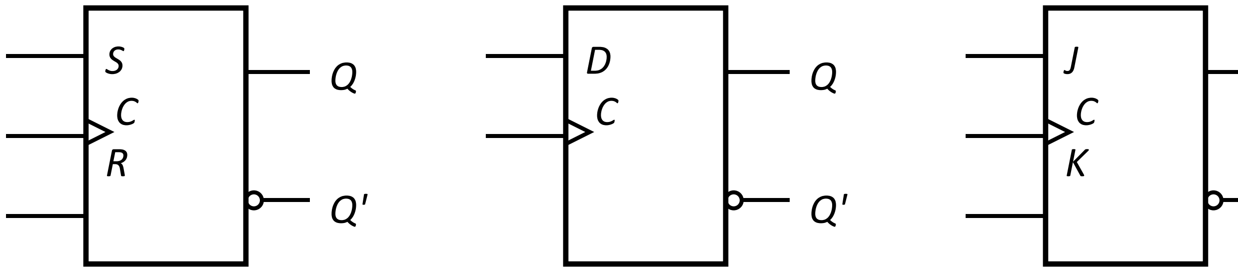
- Output changes state at a specified point on a triggering input called the *clock*.

- Change state either at the *positive edge* (rising edge) or at the *negative edge* (*falling edge*) of the clock signal.

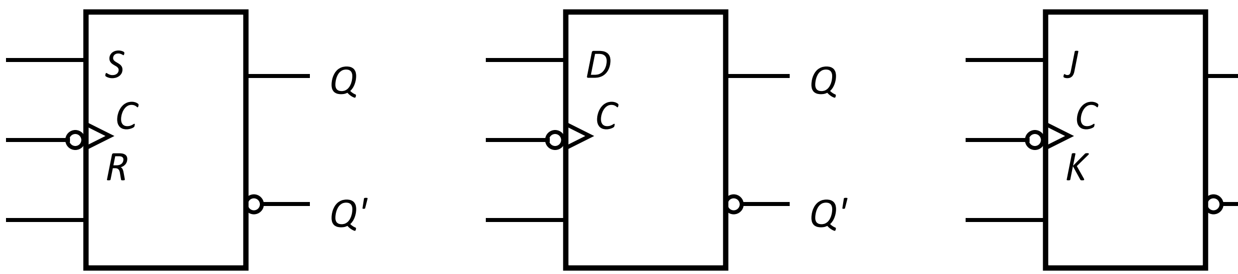


Edge-Triggered Flip-flops

- Types: S-R, D, J-K and T edge-triggered flip-flops.
- Note the “>” symbol at the clock input.



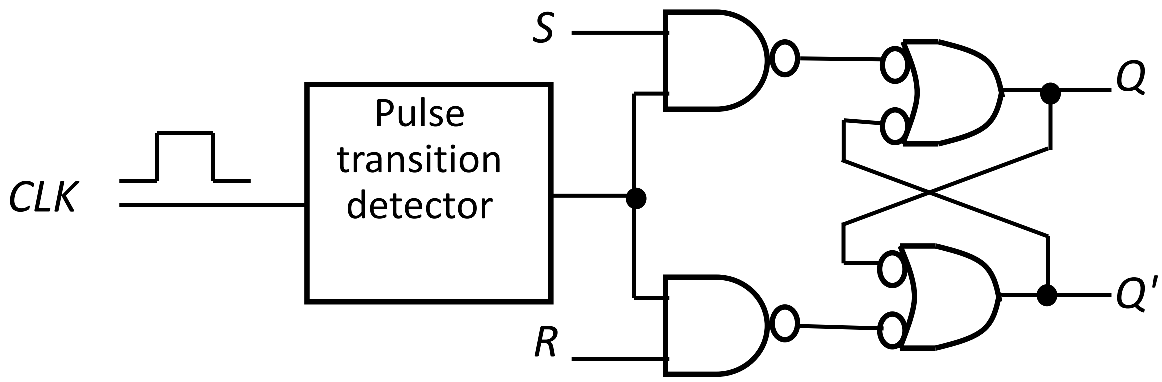
Positive edge-triggered flip-flops

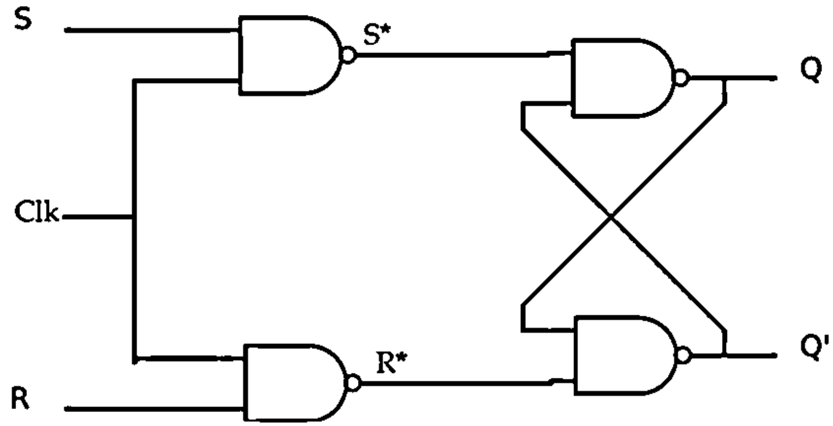


Negative edge-triggered flip-flops

S-R Flip-flop

The pulse transition detector.





S-R Flip-flop

- S-R flip-flop: on the triggering edge of the clock pulse,
  - *S*=HIGH (and *R*=LOW) a SET state
  - *R*=HIGH (and *S*=LOW) a RESET state
  - both inputs LOW a no change
  - both inputs HIGH an invalid

- Truth table of positive edge-triggered S-R flip-flop:

S	*R*	*CLK*	*Q(t+1)*	Comments
0	0	X	*Q(t)*	No change
0	1	↑	0	Reset
1	0	↑	1	Set
1	1	↑	?	Invalid

Characteristic Equation

$$Q(t+1) = S + R'.Q(t)$$

X = irrelevant ("don't care")

↑ = clock transition LOW to HIGH

Truth Table:

Clk	S	R	Q(t+1)
0	X	X	Q(t)
1	0	0	Q(t)
1	0	1	0
1	1	0	1
1	1	1	Invalid

Characteristic Table:

Clk = 1

Q(t)	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Excitation Table: Present, next, inputs

Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Characteristic Equation:

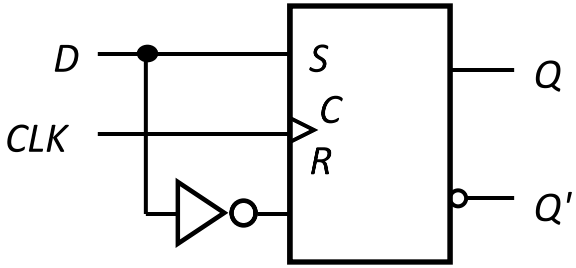
K map for Q(t+1):

Q(t) \ S R	00	01	11	10
0	0	0	X	1
1	1	0	X	1

$$Q(t+1) = S + Q(t)\,R'$$

D Flip-flop

- Single input *D* (data)
  - ❖ *D*=HIGH a SET state
  - ❖ *D*=LOW a RESET state
- *Q* follows *D* at the clock edge.
- Convert S-R flip-flop into a D flip-flop: add an inverter.



A positive edge-triggered D flip-flop formed with an S-R flip-flop.

D	*CLK*	*Q(t+1)*	Comments
1	↑	1	Set
0	↑	0	Reset

↑ = clock transition LOW to HIGH

Characteristic Equation

$$Q(t+1) = D$$

D Flip Flop

Truth Table:

Clk	D	Q(t+1)
0	X	Q(t)
1	0	0
1	1	1

Characteristic

Q(t)	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table:

Q(t)	Q(t+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

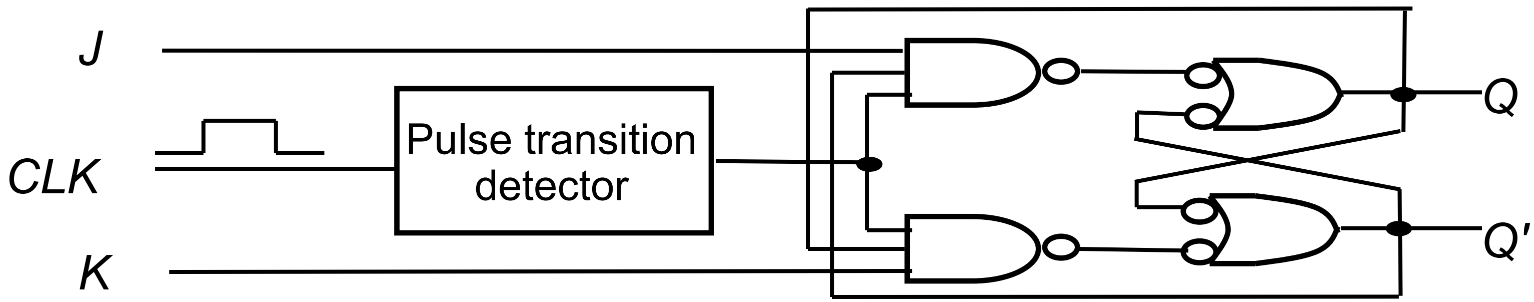
$$Q(t+1) = D$$

J-K Flip-flop (Jack Kilby)

Q and Q' are fed back to the pulse-steering NAND gates.

- No invalid state.
- Include a *toggle* state.
  - ❖ *J*=HIGH (and *K*=LOW) ⇨ SET state
  - ❖ *K*=HIGH (and *J*=LOW) ⇨ RESET state
  - ❖ both inputs LOW ⇨ no change
  - ❖ both inputs HIGH ⇨ toggle

J-K Flip-flop



- Truth table.

J	K	CLK	Q(t+1)	Comments
0	0	↑	*Q(t)*	No change
0	1	↑	0	Reset
1	0	↑	1	Set
1	1	↑	*Q(t)'*	Toggle

Characteristic Equation

$$Q(t+1) = J.Q(t)' + K'.Q(t)$$

Truth Table:

Clk	J	K	Q(t+1)
0	X	X	Q(t)
1	0	0	Q(t)
1	0	1	0
1	1	0	1
1	1	1	Q(t)’

Excitation Table:

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

K map for J:

Q(t) \ Q(t+1)	0	1
0	0	1
1	X	X

J = Q(t+1)

K map for K:

Q(t) \ Q(t+1)	0	1
0	X	X
1	1	0

K = Q(t+1)’

Characteristic Table:

Q(t)	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

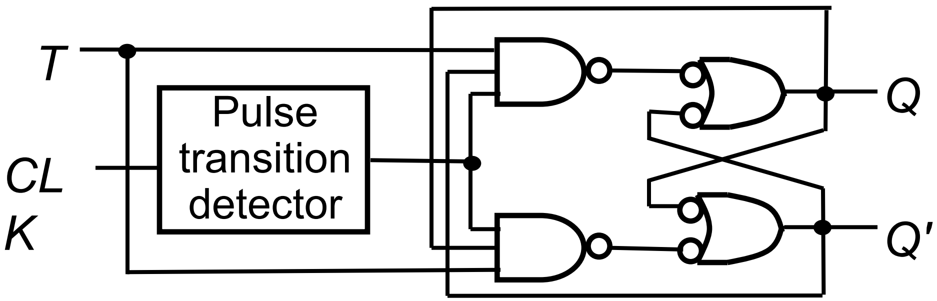
K map for Q(t+1):

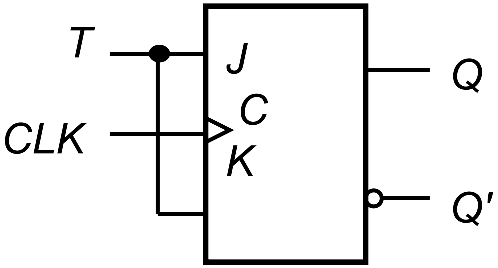
Q(t) \ JK	00	01	11	10
0	0	0	1	1
1	1	0	0	1

Q(t+1) = J.Q(t)’ + K’.Q(t)

T Flip-flop

- T flip-flop: single-input version of the J-K flip flop, formed by tying both inputs together.





- Truth table.

T	*CLK*	*Q(t+1)*	Comments
0	↑	*Q(t)*	No change
1	↑	*Q(t)'*	Toggle

Characteristic Equation

$$Q(t+1) = T.Q(t)' + T'.Q(t)$$

Truth Table:

Clk	T	Q(t+1)
0	X	Q(t)
1	0	Q(t)
1	1	Q(t)’

Characteristic Table:

Q(t)	T	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

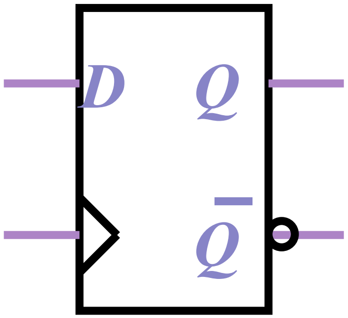
Excitation Table:

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

$$T = Q(t) \oplus Q(t+1)$$

$$Q(t+1) = Q(t) \oplus T$$

Flip-Flop Characteristic Tables and Equations

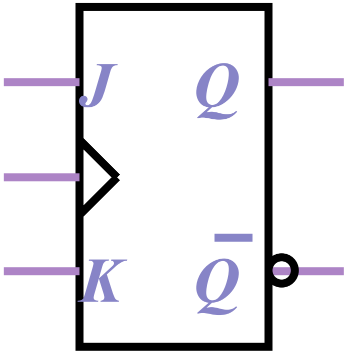


D	$Q(t+1)$
0	0
1	1

Reset

Set

$Q(t+1) = D$



J	K	$Q(t+1)$
0	0	$Q(t)$
0	1	0
1	0	1
1	1	$Q'(t)$

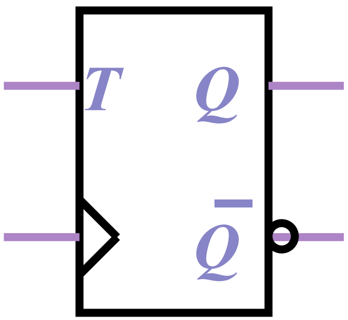
No change

Reset

Set

Toggle

$Q(t+1) = JQ' + K'Q$



T	$Q(t+1)$
0	$Q(t)$
1	$Q'(t)$

No change

Toggle

$Q(t+1) = T \oplus Q$

The excitation table for all flip-flops

Present State	Next State	SR flip-flop inputs		D flip-flop input	JK flip-flop inputs		T flip-flop input
Q	Q+1	S	R	D	J	K	T
0	0	0	x	0	0	x	0
0	1	1	0	1	1	x	1
1	0	0	1	0	x	1	1
1	1	x	0	1	x	0	0

A M-F flip-flop behaves as follows:
If MF = 01, the flip-flop changes state.
If MF = 11, the flip-flop is set to Q = 0.
If MF = 00, the flip-flop is set to Q = 1.
The input combination MF = 10 is not allowed.
(a) Give the characteristic (next-state) equation for this flip-flop.
(b) Complete the table, using don't-cares where possible.

Q	Q+	M	F
0	0		
0	1		
1	0		
1	1		

(c) Realize the following next-state equation for Q using a MF flip-flop:

$Q^+ = CQ + DQ'$. Find equations for M and F.

Flip Flop Conversions

Steps:

1. **Identify the available and required flip flop**
2. **Make the excitation table for available flip flop**
3. **Make the characteristic table for required flip flop**
4. **Write the Boolean expression for available flip flop**
5. **Draw the circuit.**

JK to D Flip Flop

1. Available FF = JK & Required FF = D
2. Excitation table for JK FF

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

3. Characteristic table for D FF:

Q(t)	D	Q(t+1)	J	K
0	0	0	0	X
0	1	1	1	X
1	0	0	X	1
1	1	1	X	0

4. Boolean expression for available JKFF:

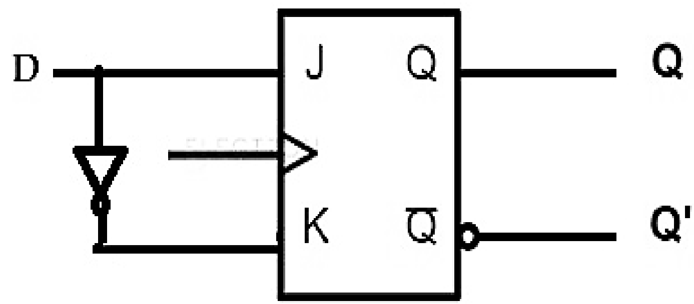
Q(t) \ D	0	1
0	0	1
1	X	X

J = D

Q(t) \ D	0	1
0	X	X
1	1	0

K = D’

5. Circuit Diagram:



SR to JK Flip Flop

1. Available FF = SR FF ;
Required FF = JK FF

2. Excitation table of SR FF:

Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

3. Characteristic table for JK FF:

Q(t)	J	K	Q(t+1)	S	R
0	0	0	0	0	X
0	0	1	0	0	X
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	1	X	0
1	0	1	0	0	1
1	1	0	1	X	0
1	1	1	0	0	1

4. Boolean expression for available SR FF:

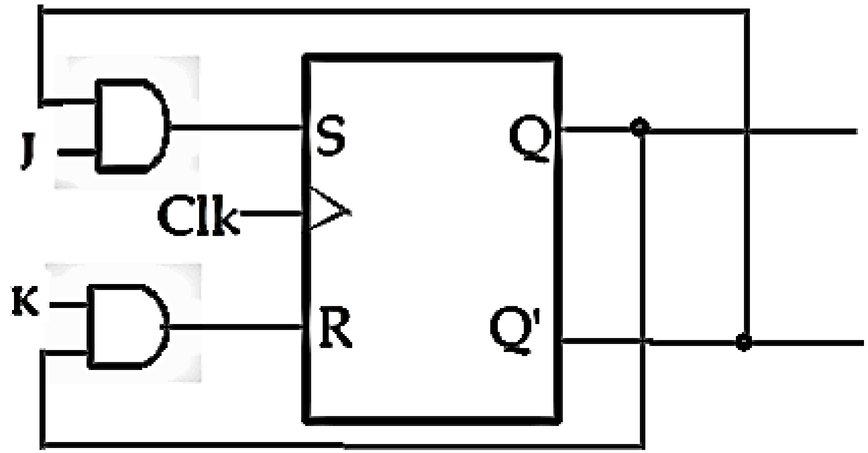
Q(t) \ J K	00	01	11	10
0	0	0	1	1
1	X	0	0	X

$$S = J\,Q(t)'$$

Q(t) \ J K	00	01	11	10
0	X	X	0	0
1	0	1	1	0

$$R = K\,Q(t)$$

5. Circuit Diagram:



SR Flip-Flop to D Flip Flop

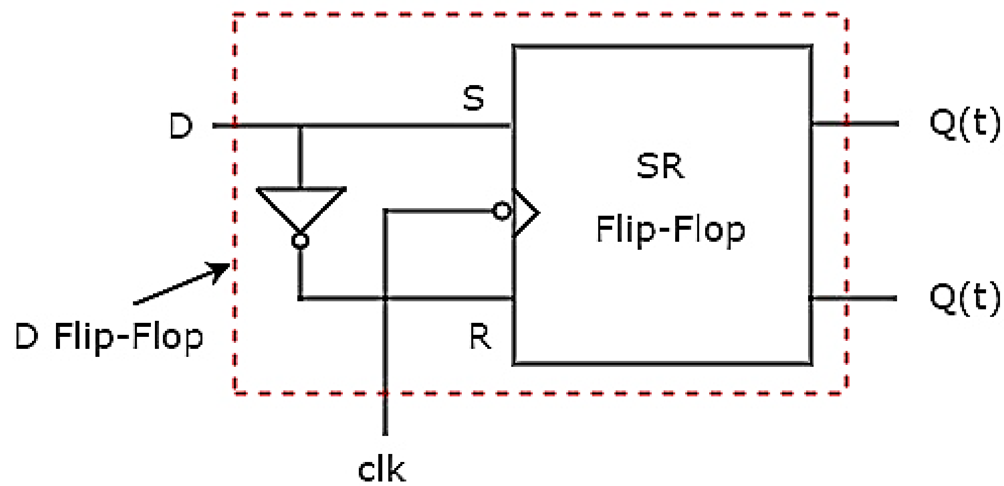
Present State	D flip-flop input	Next State	SR flip-flop inputs	
Q	D	Q_{+1}	S	R
0	0	0	0	x
0	1	1	1	0
1	0	0	0	1
1	1	1	x	0

Q(t) \ D	0	1
0	0	1
1	0	X

S = D

Q(t) \ D	0	1
0	X	0
1	1	0

R = D’



JK Flip-Flop to T Flip Flop

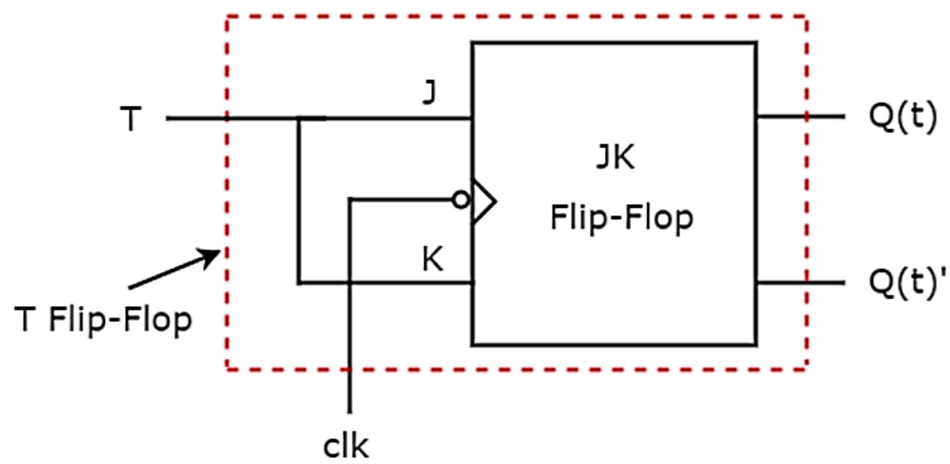
Present State	D flip-flop input	Next State	SR flip-flop inputs	
Q	T	Q_{+1}	J	K
0	0	0	0	X
0	1	1	1	X
1	0	1	X	0
1	1	0	X	1

Q(t) \ T	0	1
0	0	1
1	X	X

J = T

Q(t) \ T	0	1
0	X	X
1	0	1

K = T



JK to D Flip Flop

1. Available FF = JK &
Required FF = D

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

3. Characteristic table

Q(t)	D	Q(t+1)	J	K
0	0	0	0	X
0	1	1	1	X
1	0	0	X	1
1	1	1	X	0

4. Boolean expression for available JKFF:

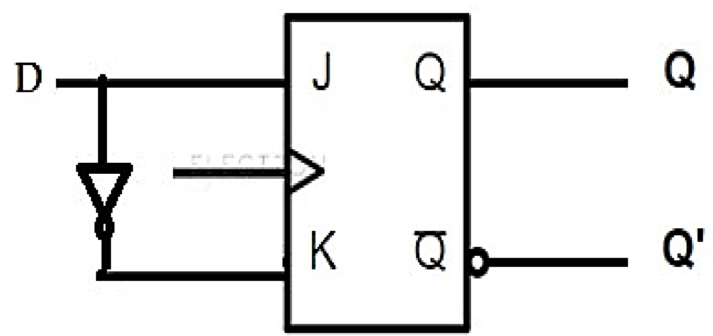
Q(t) \ D	0	1
0	0	1
1	X	X

J = D

Q(t) \ D	0	1
0	X	X
1	1	0

K = D’

5. Circuit Diagram:



JK to SR Flip Flop

S-R Inputs		Outputs		J-K Inputs	
S	R	Qp	Qp+1	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	Invalid		Dont care	
1	1	Invalid		Dont care	

JK to SR Flip Flop

S \ RQp	00	01	11	10
0	0 (0)	X (1)	X (3)	0 (2)
1	1 (4)	X (5)	X (7)	X (6)

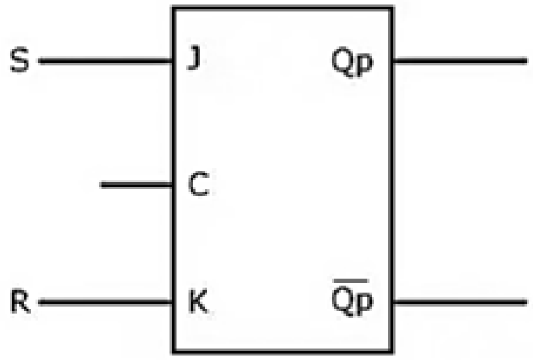
J=S

K-maps

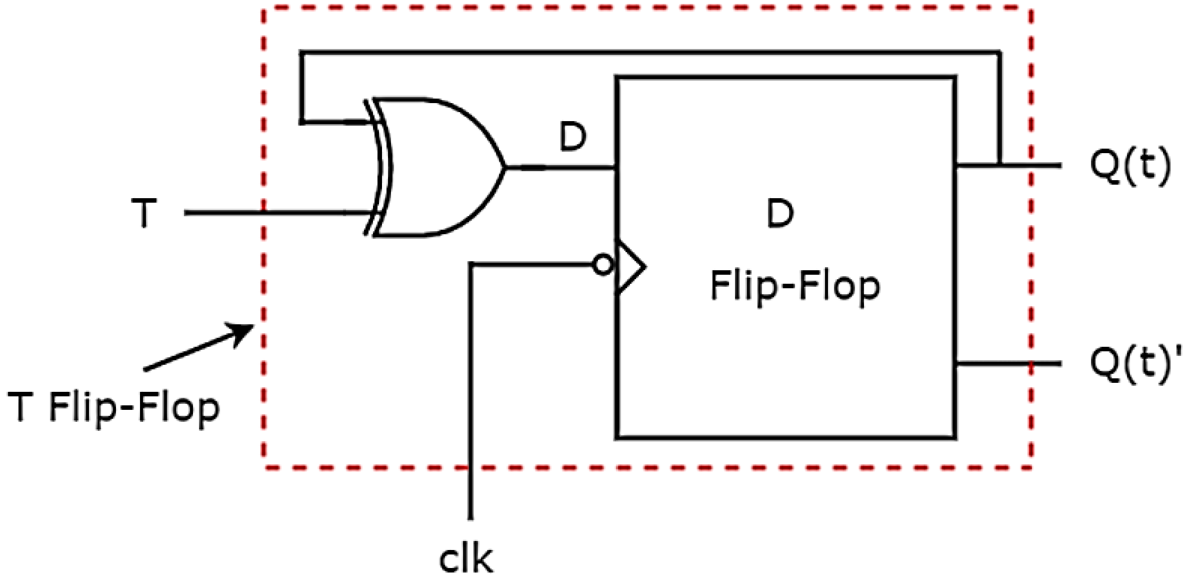
S \ RQp	00	01	11	10
0	X (0)	0 (1)	1 (3)	X (2)
1	X (4)	0 (5)	X (7)	X (6)

K=R

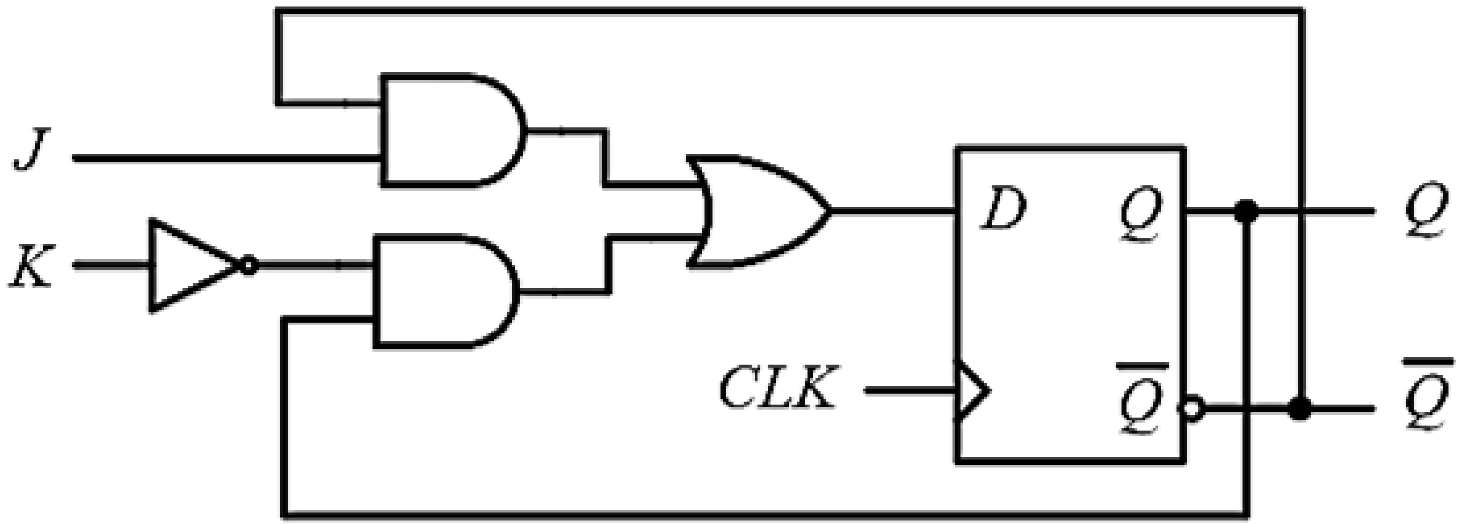
Logic Diagram



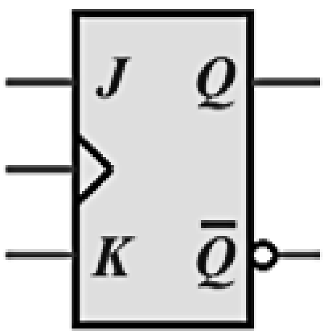
D flip-flop to T flip-flop



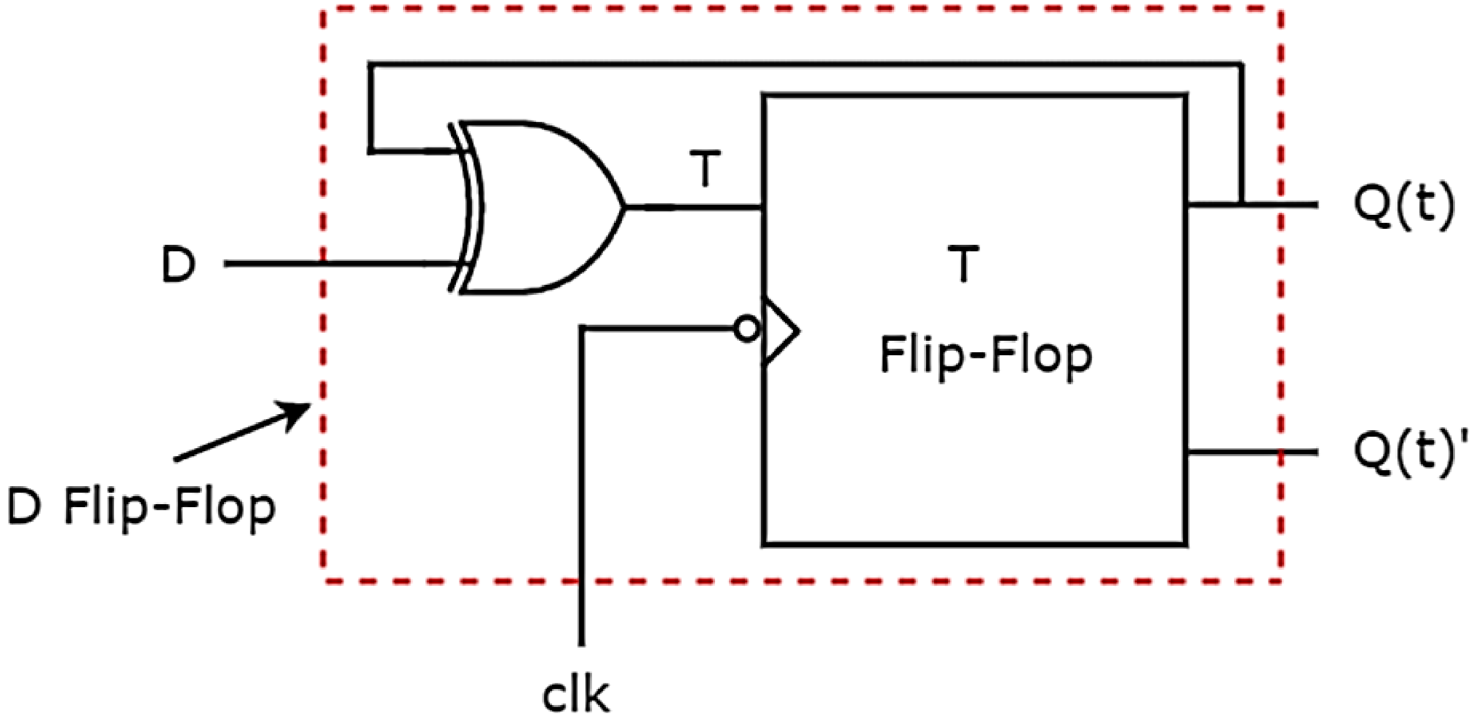
D flip flop to JK Flip Flop



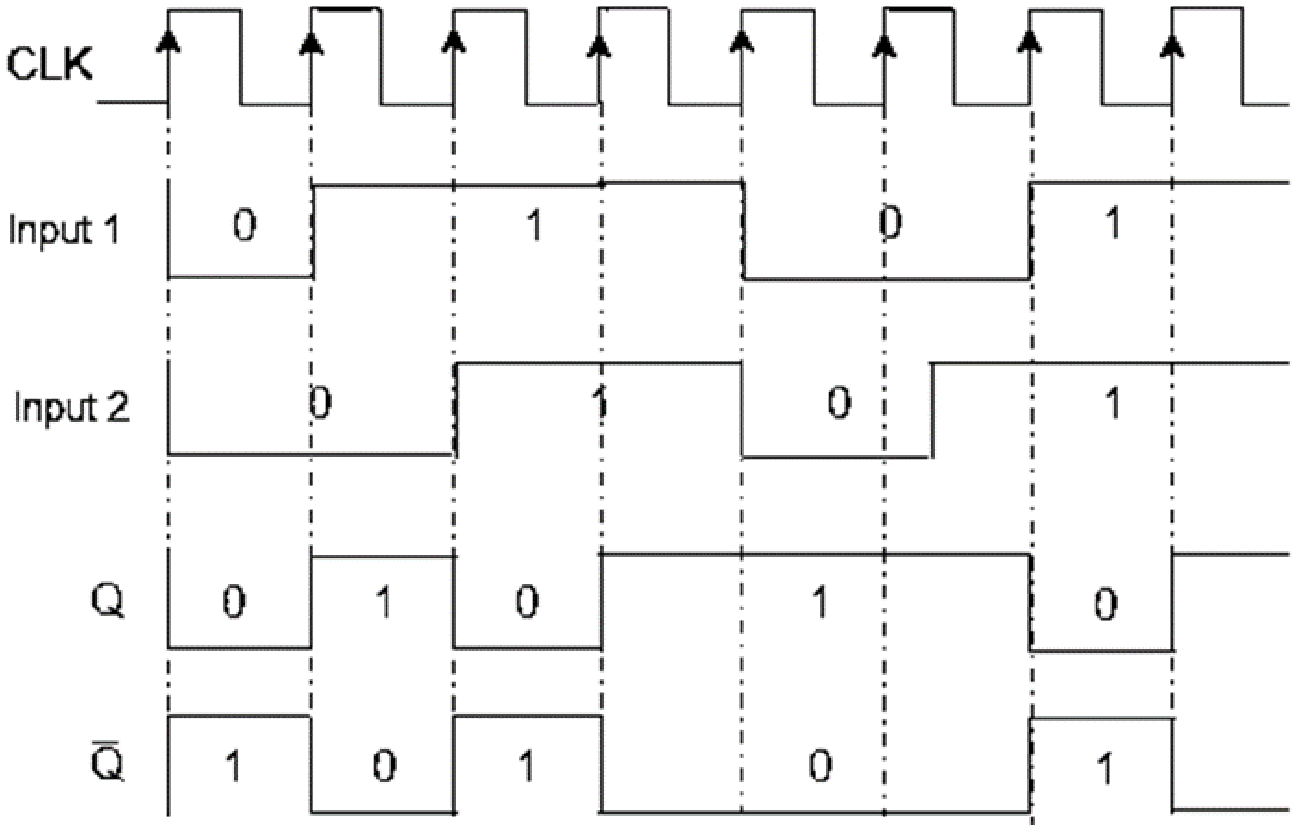
$$D = JQ' + K'Q$$



T flip-flop to D flip-flop

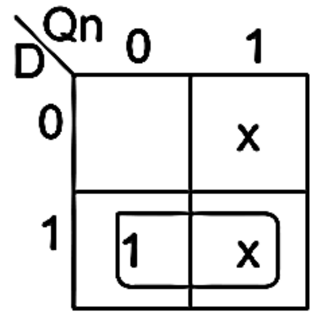


Identify the flip-flop and write its characteristic table and equation from the following timing diagram. Convert it into a D flip-flop. Explain the procedure for a flip-flop conversation in details.

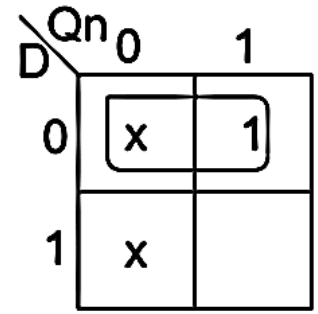


- **The given flip-flop is J K Flip-flop.**

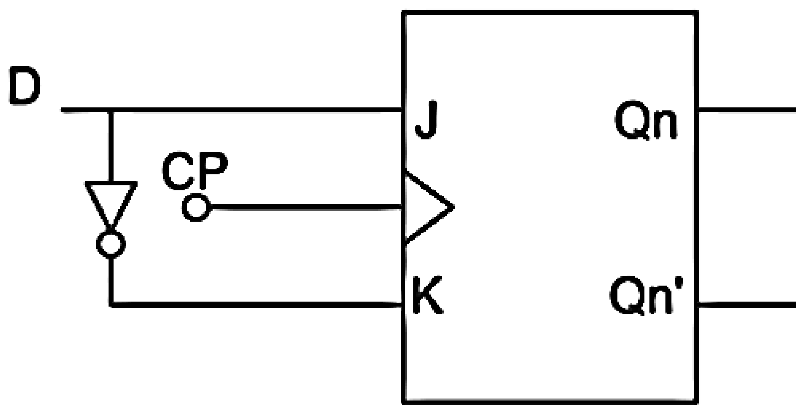
D	Qn	Qn+1	J	K
0	0	0	0	x
0	1	0	x	1
1	0	1	1	x
1	1	1	x	0



$J = D$



$K = D'$



Obtain the Boolean expressions (A>B, A=B and A<B) for 3-bit magnitude comparator. Based on the comparison, develop a control network to perform the following operation in block diagram level.

a. If A=B, Perform array multiplication

b. If A>B, Perform two complement subtraction

c. If A<B, Perform binary addition.